Marathi Dialect Normalisation & Speech Adaptation

Architectural Review, Thesis Defense & Staged Technical Blueprint

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Executive Summary & Research Crux:
This document provides the definitive architectural blueprint, theoretical thesis defense, and staged implementation plan for building a regional speech normalisation and recognition system for **Marathi dialects** (South Konkan D1, Varhadi D4, North Konkan D2 $\rightarrow$ Standard Pune D3).

It begins with a rigorous **Thesis Defense** establishing why a standalone dialect normalisation component is necessary over LoRA fine-tuning or post-ASR text cleanup. It then provides the **Marathi Speech Pipeline Review**, evaluating the 3-stage cascade (**Diarization $\rightarrow$ Dialect Normalisation $\rightarrow$ ASR**), followed by a **Staged 3-Phase Roadmap** (Phase 0: Text-to-Text MVP $\rightarrow$ Phase 1: Standalone Audio Feature Normaliser $\rightarrow$ Phase 2: Multilingual Scale-Up & ASR Adaptation Benchmark).

1 Thesis Defense: Standalone Normalisation vs. Alternatives

When defending a thesis or presenting a research proposal for a standalone dialect normalisation component (whether a full voice-conversion module or an acoustic feature adapter), academic reviewers will ask:

> *"Why build a separate normalisation component when you could fine-tune the downstream ASR model end-to-end using LoRA adapters, or clean up text transcripts after ASR?"*

This section establishes the formal academic, system-architectural, and information-theoretic case FOR a standalone normaliser and AGAINST all alternative approaches.

1.1 Thesis Core Statement

Thesis Core Defense:
We propose a **standalone, decoupled dialect normalisation component** that transforms regional non-standard Marathi speech (or its latent acoustic features) prior to downstream recognition. Standalone normalisation guarantees **universal model agnosticism**, enables **black-box API deployment**, prevents **catastrophic forgetting**, preserves **audio-level intelligibility for multi-task applications**, and disentangles **phonological shift from sequence language modeling**.

1.2 The Case FOR a Standalone Normalisation Component

1. **Decoupled Modularity & Universal Model Agnosticism (Plug-and-Play Architecture):** A standalone normaliser operates directly on speech waveforms or model-agnostic acoustic representations ($X_{\text{Dialect}} \rightarrow \hat{X}_{\text{Standard}}$). Once trained, it acts as a universal front-end that can precede **ANY downstream ASR engine**

(IndicConformer, Whisper-v3, Meta MMS, Google STT), human listeners, live translation models, or biometric speaker verification pipelines without touching downstream code or weights.

2. **Real-World Black-Box API Compatibility:** In commercial and enterprise deployments (healthcare, public infrastructure, customer support), ASR systems are black-box commercial APIs (e.g., Google Cloud STT, Azure Speech) accessed strictly over network endpoints where internal parameter fine-tuning (LoRA) is **strictly impossible**. A standalone front-end normaliser is the **only viable architectural mechanism** that enables black-box APIs to transcribe unsupported regional dialects.

3. **100% Preservation of Pre-Trained ASR Knowledge (Zero Catastrophic Forgetting):** Fine-tuning a 600M+ parameter ASR model on a small, low-resource regional dataset risks corrupting its pre-trained linguistic, semantic, and language modeling representations built on tens of thousands of hours of data. A standalone normaliser leaves the downstream ASR model 100% frozen, preserving its pre-trained robustness.

4. **Multi-Task Usability & Audio Output Retention:** Direct ASR fine-tuning destroys the audio signal, producing only text strings. A standalone audio/feature normaliser preserves or reconstructs the speech signal, enabling human cross-dialect listening (standard Pune speakers understanding regional Varhadi/Konkan speech in real time), voice dubbing, speech-to-speech translation, and emotion preservation.

5. **Disentanglement of Phonology vs. Language Modeling:** Dialect variation is fundamentally an **acoustic-phonological transformation**, not a structural semantic change. Conflating acoustic adaptation with sequence-to-sequence language modeling inside an ASR model forces the decoder to solve two distinct problems simultaneously. Disentangling normalisation into a specialized front-end isolates the phonological shift problem and allows explicit optimization for speaker identity preservation, acoustic naturalness (UTMOS), and phone contrast enhancement.

1.3 The Case AGAINST Direct LoRA Fine-Tuning of Downstream ASR

1. **Tightly Coupled Model Dependency:** LoRA adapters ($W'_{\text{ASR}} = W_{\text{ASR}} + BA$) bind the adaptation strictly to a single specific model architecture and checkpoint. Upgrading the underlying ASR model renders the LoRA weights useless, requiring costly re-training and re-collection of dialect data.

2. **Inapplicability to Black-Box Endpoints:** LoRA requires open access to internal weight matrices, rendering it un-deployable over third-party APIs or proprietary hardware accelerators.

3. **Destruction of Audio Intelligibility:** LoRA fine-tuning provides zero benefit to non-ASR tasks. It cannot produce audible speech for human cross-dialect understanding or live speech translation.

4. **Combinatorial Explosion across Sub-Dialects:** Fine-tuning via LoRA requires maintaining separate adapter modules for every sub-dialect (Varhadi, South Konkan, North Konkan, Marathwada). In contrast, a standalone normaliser uses a shared discrete acoustic codebook (e.g., TokAN VQ-VAE) to map arbitrary dialect tokens onto standard acoustic primitives.

1.4 The Case AGAINST Post-ASR Text Normalisation (NMT / LLM Rewriting)

1. **Fatal Information Loss Under CTC / Argmax Tokenization:** When non-standard Marathi dialect speech is processed by a standard ASR, acoustic frames mismatching standard distributions get misclassified into incorrect Devanagari subwords during CTC/beam-search decoding:

$$Y^* = \arg\max_{Y} P(Y \mid X_{\text{audio}})$$

The moment speech is tokenized into text Y^*, **acoustic phoneme information is irreversibly destroyed** (entropy loss under argmax projection). No downstream text model (NMT or LLM) can reliably guess whether a transcribed character was what the speaker said or a phonetically garbled mishearing.

2. **High-Confidence Hallucination Propagation:** ASR decoders output misheard tokens with high softmax confidence. Text rewriting models process these corrupted tokens as ground truth, leading to catastrophic semantic hallucinations.

1.5 Addressing Native Language Support in 22-Language Multilingual ASR

State-of-the-art multilingual ASR engines such as `ai4bharat/indic-conformer-600m-multilingual` natively support all 22 officially recognized Eighth Schedule languages of India—including Marathi (`mr`), Konkani (`kok`), Hindi (`hi`), Assamese (`as`), Bengali (`bn`), Gujarati (`gu`), Kannada (`kn`), Malayalam (`ml`), Tamil (`ta`), Telugu (`te`), and Urdu (`ur`). We explicitly clarify why native 22-language support does **not** eliminate the need for standalone dialect normalisation:

1. **Prestige vs. Regional Dialect Gap:** Multilingual ASR training corpora (e.g., IndicVoices) collect data primarily from urban, prestige standard speakers (Goan Standard Konkani or Pune Standard Marathi). When deployed on regional coastal/rural varieties (South Konkan D1, Malvani, Chitpavani, Mangalorean Konkani), native decoders experience severe acoustic mismatch (WER $>$ 30%). Official language inclusion does not equal regional dialect coverage.

2. **Dual-Baseline Benchmark Design:** Native 22-language support in IndicConformer provides a valuable empirical asset: it allows us to evaluate regional speech against **two official baselines**: (1) Native Konkani ASR decoding (`IndicConformer-Konkani`), and (2) Cross-lingual Marathi ASR decoding (`IndicConformer-Marathi`). Our standalone feature adapter (f_θ) acts as a universal front-end preceding **both** decoders across the 22-language suite, proving that acoustic feature normalisation closes the regional dialect gap regardless of which target decoder is selected.

1.6 Defense Summary Matrix

Architecture	Core Mechanism	Case FOR (Strengths)	Case AGAINST (Fatal Flaws)
Standalone Audio/Feature Normaliser (Proposed Thesis)	Pre-ASR acoustic feature or waveform transformation ($X_{\text{L2}} \rightarrow \hat{X}_{\text{L1}}$)	**Universal model agnosticism**; works with black-box APIs; zero catastrophic forgetting; preserves audio for human listening	Requires acoustic pair generation or discrete token alignment
Direct LoRA ASR Fine-Tuning	Parameter-efficient fine-tuning of ASR weights ($W_{\text{ASR}} + \Delta W$)	Parameter efficient (<1% weights fine-tuned); high performance on target model	Tightly coupled to 1 model checkpoint; **cannot work on black-box APIs**; destroys audio signal
Post-ASR Text Normalisation	Post-processing garbled text via NMT/LLM rewriting	Easy text-only pair collection	**Fatal flaw:** Cannot recover acoustic phonemes lost during ASR argmax tokenization

2 Review of the Planned Marathi Speech Pipeline

2.1 What This Pipeline Is Proposing

The core architecture proposes a modular, three-stage cascade for transcribing regional Marathi dialect speech using an existing high-resource Marathi speech-recognition (ASR) system:

1. **Stage 1 — Diarization (Speaker Segmentation & Clustering):** Processes raw, multi-speaker continuous audio streams to detect active speech boundaries and assign speaker labels, partitioning the input waveform into homogeneous single-speaker audio segments $A_1, A_2, \ldots, A_K$.

2. **Stage 2 — Audio Dialect Normalisation / Acoustic Adaptation:** Transforms the regional Marathi acoustic signal A_i directly in the signal or feature space into an adapted representation $\hat{A}_i$ calibrated to match the phonetic and acoustic expectations of the target Marathi ASR model.

3. **Stage 3 — ASR Transcription (Marathi Acoustic Model):** Feeds the adapted acoustic representation $\hat{A}_i$ into a pre-trained Marathi ASR backbone (e.g., IndicConformer-600M or IndicWhisper) to emit final Devanagari text transcripts Y_i.

2.2 Diarization: Technical Execution & Language Isolation

Diarization operates strictly on acoustic signal characteristics (pitch, formant structures, spectral envelope, speaker embeddings) and VAD (Voice Activity Detection) timestamps, independent of linguistic semantics or underlying language phonology.

- **Execution Workflow:** Raw audio passes through a neural VAD (e.g., Pyannote-Audio 3.1) to trim silence. Active speech frames are projected into a metric space using speaker encoders (e.g., ECAPA-TDNN or x-vectors). Agglomerative Hierarchical Clustering (AHC) or Variational Bayesian HMM (VBx) groups vectors into distinct speaker IDs.

- **Isolation Guarantee:** Because diarization functions purely at the acoustic identity layer, it does not interact with the dialect discrepancy or language boundary problem. Placing it first isolates downstream processing to clean single-speaker segments.

2.3 Audio Dialect Normalisation vs. Text Cleanup

Fixing speech at the audio/feature stage targets the root cause of recognition failure rather than attempting to patch downstream symptoms. When regional Marathi phoneme sequences (e.g., retroflex lateral lenition in Varhadi, mid-vowel raising in South Konkan, or dental-palatal affricate contrasts) differ from the Standard Marathi acoustic model's training distribution, the acoustic model misclassifies the phoneme frames into incorrect character tokens. Pre-ASR normalisation ($X_{\text{Dialect}} \rightarrow \hat{X}_{\text{Standard}}$) preserves phonemic contrast before irreversible tokenization occurs.

2.4 Technical Options for the Normalisation Stage

1. **Full Waveform Voice Conversion:** Converts regional Marathi speech waveforms to sound like standard Pune Marathi in the original speaker's voice using TokAN-style VQ-VAE + Autoregressive Transformer + Flow Matching Vocoder. High quality, but complex neural synthesis.

2. **Acoustic Feature Adaptation (Learned Front-End Filter):** Operates directly on intermediate 80-channel Log-Mel spectrograms or self-supervised HuBERT/wav2vec2 embeddings:

$$\hat{F}_{\text{Standard}} = f_\theta(F_{\text{Dialect}})$$

   A 2-to-4 layer Conformer/Linear feature adapter shifts regional acoustic features into the feature space expected by the Standard Marathi ASR encoder without generating audible waveforms. Fast, lightweight (<10M params).

3. **Direct ASR Fine-Tuning (LoRA Adapter):** Injects LoRA modules ($r = 16, \alpha = 32$) into the Marathi ASR model's attention/FFN layers and fine-tunes directly on regional dialect speech.

2.5 Overall Verdict on the Pipeline

Strategic Verdict & Technical Path: Keep the 3-stage order (**Diarization** $\rightarrow$ **Dialect Normalisation** $\rightarrow$ **ASR**). Treat the normalisation stage initially as a lightweight, well-scoped acoustic feature adapter, and plan from day one to evaluate direct ASR fine-tuning as an empirical baseline.

3 Staged 3-Phase Execution Roadmap

To de-risk development, optimize GPU compute budget on a single 32GB GPU, and empirically establish performance bounds, we adopt a strict **staged 3-phase execution strategy**:

3.1 Phase 0: Text-to-Text Normalisation MVP & Parallel Corpus Engine (Month 1)

Goal: Build and evaluate a discrete Text-to-Text Normalisation model ($T_{\text{Dialect}} \rightarrow T_{\text{Standard}}$) **FIRST**, before touching complex acoustic pipelines.

3.1.1 1. Strategic Purpose of Phase 0

Starting with Text-to-Text normalisation serves three vital research objectives:

1. **Establishes Upper-Bound Benchmark:** Evaluates the maximum achievable translation/normalisation accuracy on *clean ground-truth regional transcripts* vs. *garbled ASR transcripts*.

2. **Generates Parallel Data for Voice Cloning (Phase 1 Driver):** Creates thousands of aligned parallel sentence pairs ($T_{\text{Standard}}, T_{\text{Dialect}}$). These aligned text pairs directly drive the Phase 1 **IndicF5** voice-cloning TTS engine to generate parallel acoustic audio pairs ($X_{\text{L1\_Real}}, X_{\text{L2\_Synthetic}}$).

3. **Empirical Proof of Acoustic Loss Hypothesis:** Demonstrates that text rewriting models (NMT/LLM) fail when applied post-ASR due to CTC/argmax phonetic corruption, providing the empirical justification for the Phase 1 pre-ASR feature adapter.

3.1.2 2. Candidate Model Backbones & Hyperparameters

We evaluate three model families on a single 32GB GPU:

- **IndicTrans2-1B Seq2Seq (`ai4bharat/indictrans2-indic-indic-1B`):** Specialized NMT backbone trained across Devanagari Indic languages. Fine-tuned with LoRA ($r = 16, \alpha = 32$) on target Marathi dialect pairs.

- **mT5-Base / mT5-Large (`google/mt5-base`):** Multilingual T5 encoder-decoder fine-tuned using sequence-to-sequence cross-entropy loss.
- **Llama-3.1-8B-Instruct (`meta-llama/Meta-Llama-3.1-8B-Instruct`):** Causal LM fine-tuned via 4-bit QLoRA ($r = 16, \alpha = 32$, target modules: `q_proj, v_proj, k_proj, o_proj`) using Unsloth/TRL.

3.1.3 3. Parallel Corpus Construction Pipeline

Because large parallel dialect text corpora do not exist off-the-shelf, we construct a 30,000-sentence parallel text dataset using a 3-tier strategy:

1. **Tier 1 — RESPIN Marathi Transcripts (Ground Truth):** 2,170+ RESPIN sentences categorized by dialect (D1 South Konkan, D2 North Konkan, D3 Standard Pune, D4 Varhadi).
2. **Tier 2 — Rule-Based Phonological Rewriter:** Deterministic conversion rules mapping Varhadi/Konkan orthographic markers to Standard Pune Marathi (e.g., retroflex lateral $j \rightarrow ḷ$, case postpositions *-le* $\rightarrow$ *-lā*, locative markers *-at* $\rightarrow$ *-āt*).
3. **Tier 3 — LLM Synthetic Dialect Expansion:** Prompting Frontier LLMs (e.g., Llama-3.1-70B / Claude-3.5-Sonnet) with dialect lexicons and phonological rules to paraphrase 25,000 Standard Marathi sentences into authentic Varhadi (T_{Varhadi}) and South Konkan (T_{Konkan}) phrasing.

3.1.4 4. Week-by-Week Execution Blueprint (”What To Do”)

1. **Week 1 (Rule Base & Lexicon Mining):** Implement Python rule-based phonological rewriter; extract 5,000 parallel words/phrases from Marathi dialect dictionaries; build baseline text preprocessing scripts.
2. **Week 2 (Data Expansion & LLM Prompting):** Run LLM synthetic expansion script to produce 30,000 aligned ($T_{\text{Standard}}, T_{\text{Dialect}}$) sentence pairs; execute quality filtering (BLEU, length ratio verification).
3. **Week 3 (Model Fine-Tuning):** Fine-tune `IndicTrans2-1B` and `Llama-3.1-8B` QLoRA on a 32GB GPU (bfloat16, learning rate 2×10^{-4}, batch size 16, 5 epochs, $\sim$4 hours training time).
4. **Week 4 (Evaluation & Decision Gate):**
   - Evaluate BLEU, chrF++, CER, and WER on clean ground-truth text.
   - Feed raw, garbled ASR outputs from IndicConformer into the fine-tuned text model; measure accuracy drop.
   - Export clean parallel text pairs to launch Phase 1 IndicF5 voice-cloning synthesis.

3.2 Phase 1: Standalone Audio Feature Normaliser (Months 2–4, Paper 1)

Goal: Build the pre-ASR acoustic normaliser ($F_{\text{Dialect}} \rightarrow \hat{F}_{\text{Standard}}$ or TokAN discrete VQ-VAE token converter) operating on raw speech or Log-Mel features, conditioned on ECAPA-TDNN speaker embeddings.

- **Primary Pairs:** Varhadi (D4) & South Konkan (D1) $\rightarrow$ Standard Pune (D3 / IndicVoices-R Marathi).
- **Synthetic Data Pipeline:** Use IndicF5 voice-cloning TTS to generate synthetic L2 audio in real L1 speaker voices from Phase 0 parallel text.
- **Target Submission:** INTERSPEECH 2027 or ICASSP 2027 (system/results paper).

3.3 Phase 2: Multilingual Scale-Up & ASR Adaptation Benchmark (Months 5–8, Paper 2)

Goal: Expand normalisation across multiple regional dialects (South Konkan, Varhadi, Awadhi) and evaluate direct ASR acoustic model fine-tuning as an empirical comparator.

- **Extended Pairs:** Hindi Awadhi Bundeli $\rightarrow$ Khariboli (RESPIN Hindi D3 $\rightarrow$ D1).
- **Multilingual Model:** Shared VQ codebook + language/dialect-ID conditioning across Marathi and Hindi.
- **Target Submission:** IEEE/ACM TASLP or Computer Speech and Language (CSL journal).

4 Core Model Architecture: Acoustic Feature Adaptation Engine (f_θ)

The primary architectural focus of this project is the **Learned Neural Acoustic Feature Adapter** (f_θ). Instead of performing full audio-to-audio waveform generation (which requires heavy vocoders, flow matching, and phase alignment), f_θ operates as a lightweight, fast front-end filter directly in the latent acoustic representation space:

$$\hat{F}_{\text{Standard}} = f_\theta(F_{\text{Dialect}})$$

It projects non-standard regional Marathi acoustic vectors into the exact representation manifold expected by the frozen downstream Standard Marathi ASR model (`ai4bharat/indicconformer_stt_mr_hybrid_ctc_rnnt_large` or `IndicWhisper`).

4.1 Feature Space Representation Options

The feature adapter f_θ can be configured across two primary representation spaces:

1. **80-Channel Log-Mel Filterbanks ($F_{\text{Mel}} \in \mathbb{R}^{T\times 80}$):** Processes standard acoustic spectrogram frames sampled at 16 kHz with a 25 ms window and 10 ms hop size. f_θ transforms regional spectral energy distributions (e.g., formant shifts, retroflex lenition, vowel raising) directly before feature projection into the ASR encoder.
2. **Self-Supervised Speech Representations ($F_{\text{SSL}} \in \mathbb{R}^{T'\times D}$):** Extracts frame-level embeddings from intermediate layers of a frozen self-supervised speech encoder (e.g., `wav2vec2-large-xlsr-53` or `IndicHuBERT`, $D = 768/1024$). f_θ maps regional latent vectors into standard phonetic clusters.

4.2 Neural Adapter Networks

We evaluate two lightweight (<10M parameter) architecture variants for f_θ:

4.2.1 1. Conformer Feature Adapter (2-to-4 Layers)

Composed of 2-to-4 stacked Conformer blocks (Multi-Head Self-Attention + Depthwise Separable Convolution + Feed-Forward Residual Modules):

- **Depthwise Separable Convolutions:** Capture local spectral transitions, shift formants, and align phoneme duration boundaries.
- **Self-Attention Blocks:** Capture global context across the sentence, resolving non-local grammatical and postposition shifts.
- **Parameter Footprint:** $\sim$4.8M parameters ($D_{\text{model}} = 256$, 4 heads, $D_{\text{ffn}} = 1024$).

4.2.2 2. Residual Linear Spectrogram Adapter

A 4-layer 1D Convolutional Residual Network with Layer Normalization and Swish activations:

- **Residual Connections:** $\hat{F}_l = F_{l-1} + \text{Conv1D}(\text{LayerNorm}(F_{l-1}))$. Ensures stable gradient flow and prevents over-smoothing of fine spectral details.
- **Parameter Footprint:** $\sim$1.2M parameters. Ultra-fast inference with virtually zero latency.

4.3 Joint Optimization & Loss Functions

The feature adapter f_θ is trained end-to-end using a compound loss function:

$$\mathcal{L}_{\text{adapter}} = \alpha \cdot \mathcal{L}_{\text{ASR\_CTC}} + \beta \cdot \mathcal{L}_{\text{Feature\_Dist}} + \gamma \cdot \mathcal{L}_{\text{Cosine}}$$

1. **Downstream ASR CTC/Transducer Loss ($\mathcal{L}_{\text{ASR\_CTC}}$):** Passed directly from the frozen downstream ASR model's CTC loss function on target ground-truth transcripts:

$$\mathcal{L}_{\text{ASR\_CTC}} = \text{CTC\_Loss}\Big(\text{ASR\_Encoder}\big(f_\theta(F_{\text{Dialect}})\big), Y_{\text{Standard\_Text}}\Big)$$

   This forces the adapter to optimize directly for downstream ASR transcription accuracy without altering ASR weights.

2. **Parallel Feature Distance Loss ($\mathcal{L}_{\text{Feature\_Dist}}$):** Measures L_1 frame-level error against synthetic/real parallel standard frames:

$$\mathcal{L}_{\text{Feature\_Dist}} = \left\| f_\theta(F_{\text{Dialect}}) - F_{\text{Standard\_Real}} \right\|_1$$

3. **Cosine Feature Direction Loss ($\mathcal{L}_{\text{Cosine}}$):** Aligns feature vector trajectories in embedding space: $1 - \cos(\hat{F}, F_{\text{Real}})$.

4.4 TokAN Discrete Token Baseline (Comparative Hybrid)

As a discrete baseline, we adapt TokAN [2] using a discrete VQ-VAE tokenizer (512 codes, 256 dimensions) + 6-layer AR Transformer + ECAPA-TDNN speaker embedding conditioning. The comparative experiment evaluates whether continuous feature adaptation (f_θ) outperforms discrete acoustic token mapping ($T_{\text{L2}} \rightarrow T_{\text{L1}}$) under modest compute constraints.

5 Marathi Dialect Landscapes & Phonological Shift Rules

5.1 Primary Target: Marathi Regional Dialect Spectrum

Marathi exhibits rich regional dialectal variation across Maharashtra. Standard Marathi (Puneri / Pune variety) serves as the primary ASR target language:

Dialect / Variety	Region	Key Phonological Differences from Standard
Standard (Pune, D3)	Pune / Urban MH	Standard prestige variety — primary benchmark target

Dialect / Variety	Region	Key Phonological Differences from Standard
South Konkan (D1)	Sindhudurg, Ratnagiri	Coastal phonology, vowel shifts, mid-vowel raising, unique affixes
Varhadi (D4)	Vidarbha (E. MH)	Retroflex lateral [ɭ] → [j], *-le* case marker (vs *-lā*), ergative alignment
North Konkan (D2)	Palghar, Thane	Distinct palatal affricates, nasalisation loss, vocabulary overlap
Awadhi (Hindi D3)	Central UP	Dative /ka:/ → /ko:/, locative /ma:/ → /mẽ:/, aspirated affricate mergers

5.2 Concrete Sound Rules

- **Varhadi → Standard Pune:** Retroflex lateral lenition ([ɭ] → [j]), dative case marker change (*-le* → *-lā*), de-palatalisation of dental/palatal affricates.
- **South Konkan → Standard Pune:** Schwa syncope reduction, mid-vowel raising (/o/ → [u], /e/ → [i]), coastal postposition alignment.
- **Awadhi → Khariboli (Hindi):** Postposition shifts (/ka:/ → /ko:/, /ma:/ → /mẽ:/), retroflex stop lenition.

6 Data Strategy & Synthetic Parallel Data Pipeline

6.1 Available Speech Corpora

Language	Dataset	Hours	Dialect Labels	Access
Marathi	RESPIN-S1.0 [12]	∼1,200	4 dialects (Konkan South, North, Pune, Varhadi)	Direct (GitHub)
	OpenSLR 103 [9]	99	3 sociolect groups	Direct
	IndicVoices-R [6]	100+	Studio-quality standard Marathi	HuggingFace
Hindi	RESPIN-S1.0 [12]	∼2,500	5 dialects (Khariboli, Awadhi, Braj, Marwari, Garhwali)	Direct (GitHub)
	LAHAJA [19]	12.5	83 districts (test benchmark)	HuggingFace
	IndicVoices-R [6]	175+	Standard Khariboli reference	HuggingFace

6.2 The Source-Synthesis Data Pipeline (CosyAccent Methodology)

Supervised normalisation models require parallel speech pairs $(X_{\mathrm{L2}}, X_{\mathrm{L1}})$ where the same speaker reads identical sentences in both dialect and standard varieties. Because no parallel human recordings exist for Indic dialects, we adopt the **CosyAccent** [3] synthetic generation strategy:

1. **Keep L1 Target Real:** Standard audio recordings from IndicVoices-R Marathi / OpenSLR 103 ($\sim$5,000 utterances) are kept as authentic human speech. Synthesising L1 targets is strictly avoided to prevent propagating TTS artifacts into the normaliser.
2. **Synthesise L2 Dialect Audio in Cloned Voice:** Using a voice-cloning TTS (**IndicF5** [23]), we feed the L1 speaker's real voice clip as identity reference alongside Phase 0 parallel text, generating synthetic L2 speech in the **same speaker's voice**.
3. **Quality Filtering Pipeline:**
   - *ASR Consistency:* Transcribe L1 and L2 with IndicWhisper; discard if WER $>$ 15%.
   - *Speaker Similarity:* ECAPA-TDNN cosine similarity between real L1 and synthetic L2 audio ($>$ 0.5).
   - *Naturalness:* UTMOS score filter ($>$ 3.0).

6.3 IndicF5 Voice-Cloning Design Fork

Design Fork Decision Gate (Week 1 Sanity Check):

- **Path 1 (LoRA Accent Adapter):** Train a LoRA adapter on RESPIN dialect audio without modifying IndicF5. Conditioned on Speaker X's reference clip + dialect LoRA. If timbre remains anchored to Speaker X while pronunciation changes to Varhadi/Awadhi, Path 1 succeeds.
- **Path 2 (VANI-Style Disentangled Conditioning):** Redesign IndicF5 to accept separate Speaker Embeddings (ECAPA-TDNN) and Accent Embeddings. Pursued if Path 1 causes speaker timbre drift.

6.4 32GB GPU Memory Budget

Component	Configuration & Approach	VRAM (32GB)
Phase 0 Text Engine	IndicTrans2-1B / Llama-3-8B (QLoRA, bfloat16)	$\sim$8 GB
L1 TTS Voice Cloning	IndicF5 (0.4B flow-matching, reference audio)	$\sim$12 GB
VQ Tokenizer	512 codebook size, 256 embedding dimension	$\sim$4 GB
AR Token Converter	6-layer Transformer (512 hidden, 8 heads)	$\sim$6 GB
ECAPA Speaker Encoder	Frozen pre-trained PyTorch checkpoint	$\sim$2 GB
GRPO RL Fine-Tuning	LoRA adapters, micro-batch size 4	$\sim$8 GB

7 Evaluation Benchmark & Publication Strategy

7.1 Evaluation Metrics & Baselines

- **Word Error Rate (WER) & Relative WER Reduction:** Transcribe raw L2 vs converted audio using IndicConformer-600M / IndicWhisper. Target: $>$ 30% relative WER reduction.
- **Speaker Similarity:** ECAPA-TDNN cosine similarity between input L2 and converted audio ($>$ 0.85).

- **Audio Quality:** UTMOS naturalness score (> 3.0).
- **Dialect Classifier Score:** CNN trained on HuBERT features from RESPIN. Target: $> 80\%$ classification as ”standard” post-conversion.
- **Baselines:** (1) Raw un-converted L2 audio, (2) Cascaded ASR $\rightarrow$ TTS (transcribe then re-speak), (3) Frame-to-frame GAN, (4) Direct Spectrogram Flow Matching.

7.2 Publication Strategy (Two Independent Submissions)

1. **Paper 1 (Month 4 Submission): INTERSPEECH 2027** or **ICASSP 2027**. First Indic dialect normalisation system, synthetic parallel data pipeline, RESPIN/LAHAJA benchmark, and single-language Marathi proof of concept.
2. **Paper 2 (Month 8 Submission): IEEE/ACM TASLP** or **Computer Speech and Language (CSL)**. Multilingual scale-up (Marathi + Hindi), cross-lingual transfer analysis, and comparison against direct LoRA ASR fine-tuning.

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